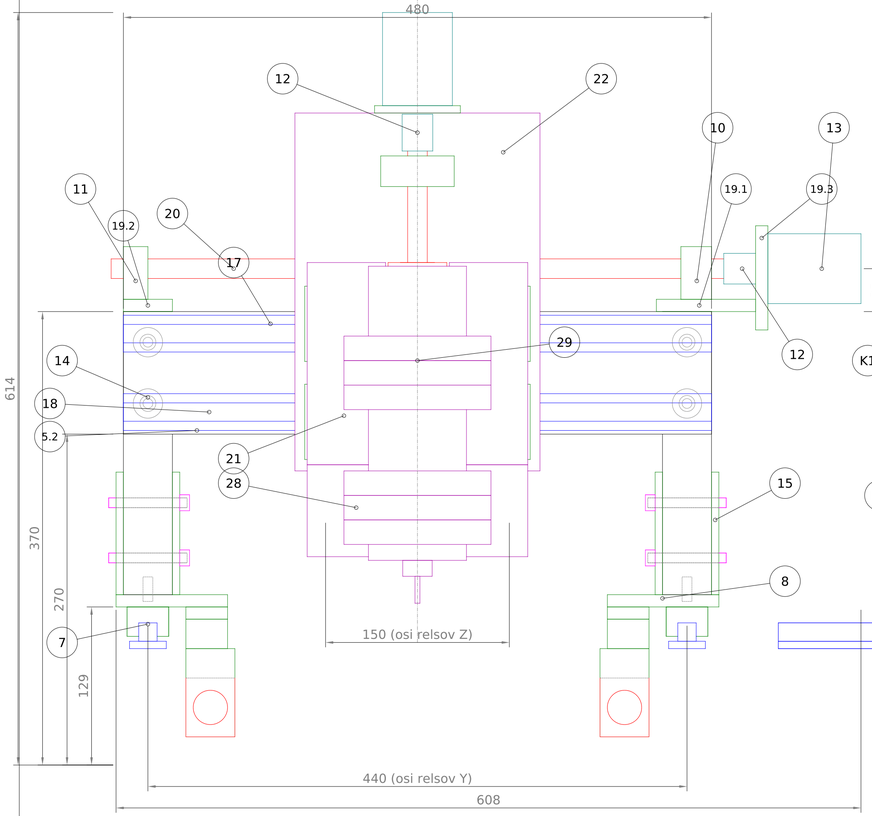
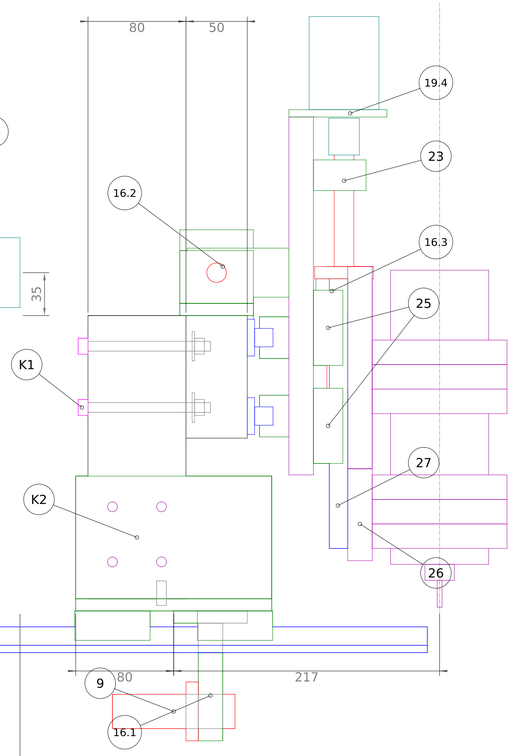
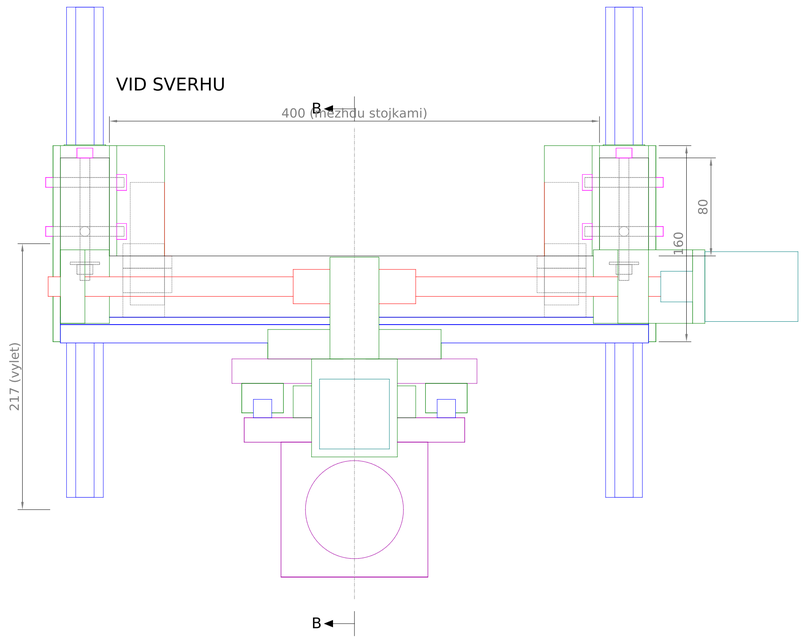
VID SPEREDI



VID SLEVA

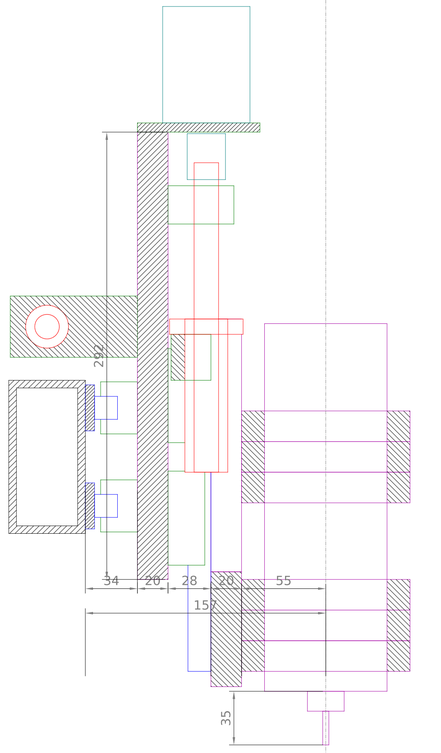


VID SVERHU



B-B (1:2)

stojki, plity 8 ne pokazany



kontsy instrumenta - na urovne stola (Z niz)

Poz.	Naimenovanie	Razmer	Kol.	Primech.
5.2	Polosa X, stal 45	480x30x6	2	list 2
7	Karetka HGH15CA (os Y)		4	pokup.
8	Plita opornaya, stal (svar. s 15)	91x160x10	2	1 + 1 zerk.
9	Gajka SHVP Y DFU1605 (vint - chert. bazy)		2	pokup.
10	Opora BK12 (X, Z)		2	pokup.
11	Opora BF12 (X)		1	pokup.
12	Mufta 6,35x10 (X, Z)	D25x30	2	pokup.
13	Motor NEMA 23, 76 mm (X, Z)		2	pokup.
14	Stojka, profil 40x80 paz 8	L=231	2	list 2
15	Scheka, stal	160x100x6	4	list 3
16.1	Kronshtejn gajki Y, svarnoj, stal	40x60x106	2	1 + 1 zerk.
16.2	Kronshtejn gajki X, Al (kvadrat 40x40)	40x83x40	1	list 3
16.3	Kronshtejn gajki Z, Al	100x26x30	1	list 3
17	Balka X, truba Al 100x50x5	L=480	1	list 2
18	Rels HGR15 (os X)	L=480	2	pokup., E=30
19.1	Plita BK12 X, Al	81x60x10	1	list 3
19.2	Plita BF12 X, Al	40x60x10	1	list 3
19.3	Plita motora X, Al	60x85x10	1	list 3
19.4	Plita motora Z, stal	70x80x6	1	list 3
20	SHVP DFU1605 C7 (os X)	L~520	1	pokup.
21	Karetka HGH15CA (os X)		4	pokup.
22	Support X, Al	200x292x20	1	list 2
23	SHVP DFU1605 C7 (os Z)	L~205	1	pokup.
25	Karetka HGH15CA (os Z)		4	pokup.
26	Plita Z, Al	180x240x20	1	list 2
27	Rels HGR15 (os Z)	L=220	2	pokup., E=20
28	Kolco homuta shpindelya D80, Al 20	120x110x20	6	list 3
29	Shpindel 1,5 kVt ER11, D80		1	pokup.
K1	Vint M8x100 DIN 912 + gajka DIN 985 + shajba DIN 9021		4	stojka - balka
K2	Vint M8x65 DIN 912 + gajka DIN 985		8	scheki - stojka
K3	Vint M8x20 DIN 7984 (nizk. golovka)		2	plita 8 - stojka
K4	Vint M4x10 DIN 912		16	karetki Y
K5	Vint M4x16 DIN 912		32	karetki X, Z
K6	Vint M4x16 DIN 912		24	relsy X, Z
K7	Vint M5x12 DIN 7991		24	polosy, plity 19
K8	Vint M5x16 DIN 7991		5	plity 19.3, 19.4
K9	Vint M5 DIN 912, dlina po mestu		6	BK12/BF12
K10	Vint M5x16 / M5x12 DIN 912		4/4	motory X / Z
K11	Vint M5x16 DIN 912		15	flantsy gaek
K12	Vint M6x20 DIN 912		8	kronshtejny 16.1
K13	Vint M6x35 DIN 912		12	homuty 28
K14	Vint M6x40 DIN 912		6	styazhka homutov
K15	Vint M6x25 DIN 912		8	kronshtejny 16.2, 16.3

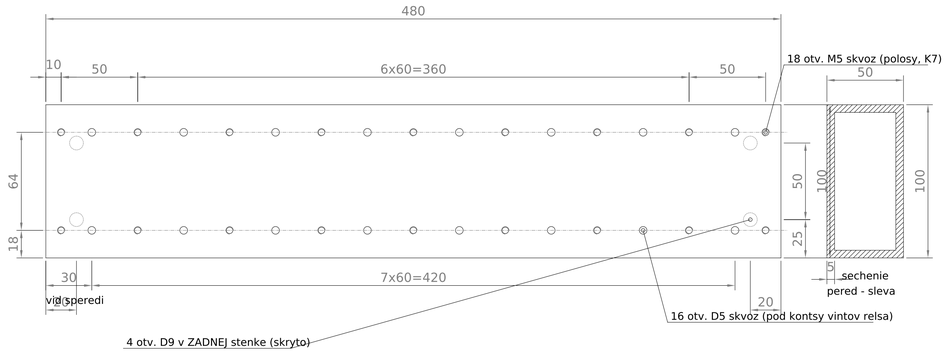
TEHNICHESKIE TREBOVANIYA

1. Razmery v mm. Polozhenie: instrument v tsentre polya, Z vnizu. Spec. v2.2. Baza so stolom - otd. chertezh.
2. Poryadok sborki:
   a) scheki 15 privarit k plite 8: snaruzhi sploshnoj shov, katet 6, kromka s razdelkoj 30x4; vnutri paza - tolko pered stojkoj (0...65 ot pereda), pod stojkoj NE varit; zatem frezerovat niz plity (ploskostnost 0,02) i paz 40 +0,1...+0,3 mezhdu schekami;
   b) plity 8 na karetki Y (K4) - DO ustanovki stoek: 4 vinta iz 8 okazhutsya pod stojkoj;
   c) stojki 14 v paz, K3 snizu; otv. D9 v stojke sverlit po schekam, K2 - 25 N*m;
   d) balka 17 na stojki (K1): gajki s shajbami DIN 9021 - vnutri truby, cherez torets (klyuch 13); perpendikulyarnost relsam Y <= 0,05 na 300 (diagonali portala), K1 - 25 N*m;
   e) polosy 5.2 (K7): obschaya ploskost <= 0,02; bazovyj rels 18 - po linejke <= 0,01/100, vtoroj parallelno <= 0,02 po indikatoru s karetki; M4 - 4 N*m;
   f) support 22: SNACHALA karetki Z (K5 s tyla), POTOM karetki X (K5 speredi) - inache ne dostat;
   g) plita Z 26: relsy 27, homuty 28 (po 3 kolca), kronshtejn 16.3; rels Z zavesti v karetki Z sverhu;
   h) BK/BF na plity 19, vint X vyverit: parallelnost relsam <= 0,05 v dvuh ploskostyah (folga pod plity); kronshtejny gaek 16 zatyagivat POSLEDNIMI, progonyaya karetku iz kraya v kraj; to zhe dlya Y i Z;
   i) shpindel: perpendikulyarnost stolu <= 0,05/100 v dvuh ploskostyah (folga pod homuty).
3. Rezba v Al: fiksator srednej prochnosti.
4. Razmery kupnyh (BK/BF, flanets DFU1605, motor, shpindel) sverit do sverlovki.

Portal s osyami X i Z. Sborka

Spec. v2.2. Detali - listy 2, 3	Masshtab 1:2,5; 1:2	List 1 / 3

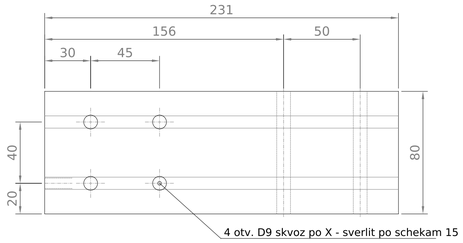
Poz. 17 Balka X, truba Al 100x50x5 (6060/6063-T6) 1 sht. M 1:2



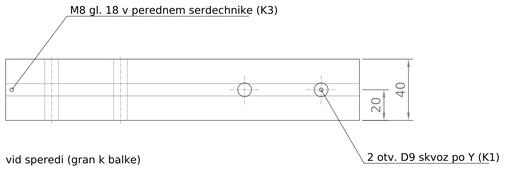


vid sverhu (pered - vnizu)

Poz. 14 Stojka, profil 40x80 paz 8 2 sht. M 1:2

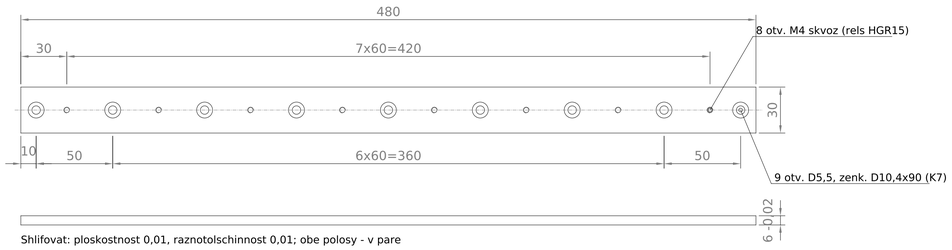


vid sboku (gran k scheke); niz - sleva, pered - vnizu



vid speredi (gran k balke)

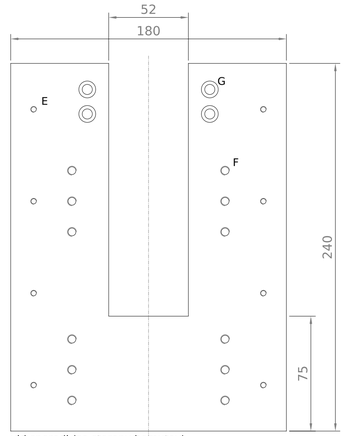
Poz. 5.2 Polosa X 30x6, stal 45 2 sht. M 1:2



Shlifovat: ploskostnost 0,01, raznotolschinnost 0,01; obe polosy - v pare

TEHNICHESKIE TREBOVANIYA K DETALYAM (LISTY 2, 3)
1. Razmery v mm, D - diametr. Neukazannye predelnye otkloneniya +-0,2; koordinaty otverstij +-0,1.
2. Plity 22, 26: litaya 5083 ili 6082-T6, obe storony frezerovat, ploskostnost 0,02, t = 20 +-0,05.
3. Truba 17: tortsy podrezat, perpendikulyarnost 0,1; otv. v perednej stenke sverlit po polose 5.2 kak po konduktoru.
4. Profil 14 - kak list 2 bazy: 6060/6063-T6, rezba M8 metchikom v serdechnike bez rassverlivaniya.
5. Plita 8 + scheki 15, kronshtejn 16.1 - St3 / S235; svarnye uzly obrabatyvat posle svarki (list 1, 2a).
6. Kromki pritupit. Pazy profilya - uslovno.

Poz. 26 Plita Z, Al 6082-T6 / 5083 1 sht. M 1:2



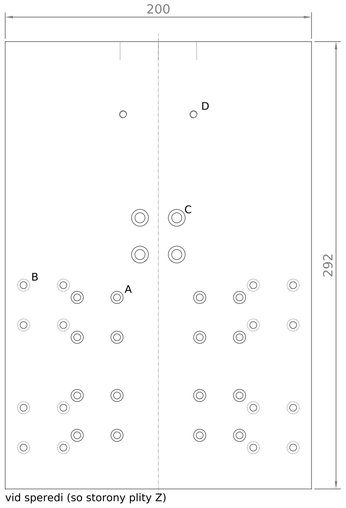
vid speredi (so storony homutov)
okno: R3 v uglah

Poz. 26: otverstiya. X - ot osi, Y - ot niza

Obozn.	Otverstie	Kol.	X	Y
E	M4 skvoz (relsy Z, K6)	8	+-75	30; 90; 150; 210
F	M6 skvoz (homuty, K13)	12	+-50	20; 40; 60; 130; 150; 170
G	D6,6, tsek. D11 gl. 6,5 speredi	4	+-40	207; 223

G - kronshtejn 16.3 (K15), s tylnoj storony

Poz. 22 Support X, Al 6082-T6 / 5083 1 sht. M 1:2



vid speredi (so storony plity Z)

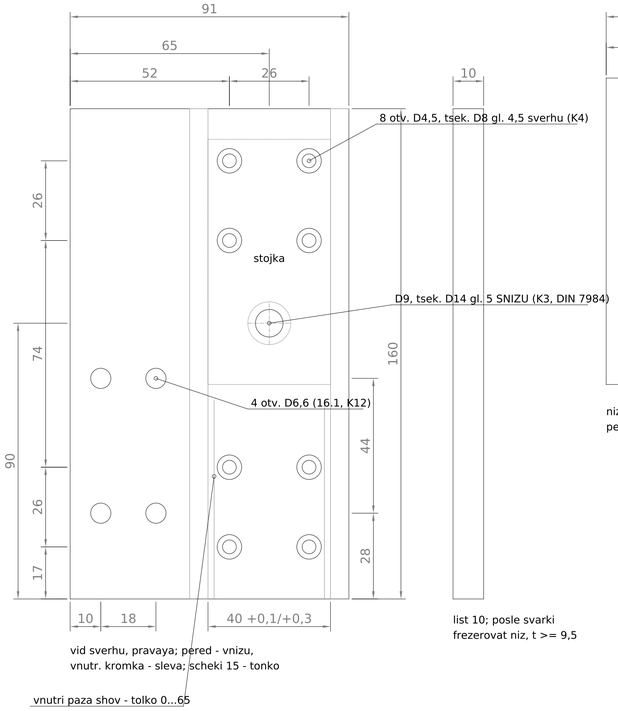
Poz. 22: otverstiya. X - ot osi, Y - ot niza

Obozn.	Otverstie	Kol.	X	Y
A	D4,5, tsek. D8 gl. 8 speredi	16	+-27; +-53	35; 61; 99; 125
B	D4,5, tsek. D8 gl. 8 SZADI	16	+-62; +-88	27; 53; 107; 133
C	D6,6, tsek. D11 gl. 6,5 speredi	4	+-12	153; 177
D	M5 gl. 12 speredi (BK12 Z)	2	+-23	244,5
E	M5 gl. 12 v verhnem tortse	3	0; +-25	po t. 10

A - karetki X (K5), B - karetki Z (K5), C - kronshtejn 16.2 (K15)

Portal. Detali			
Spec. v2.2. Sborka - list 1		Masshtab 1:2	List 2 / 3

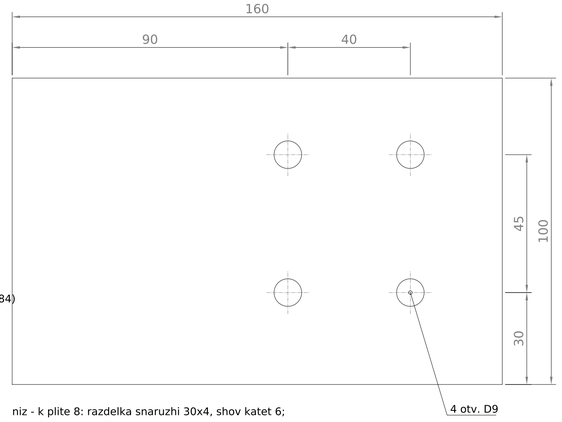
Poz. 8 Plita opornaya, stal 10 1 + 1 zerk. M 1:1



vid sverhu, pravaya; pered - vnizu,
vnutr. kromka - sleva; scheki 15 - tonko

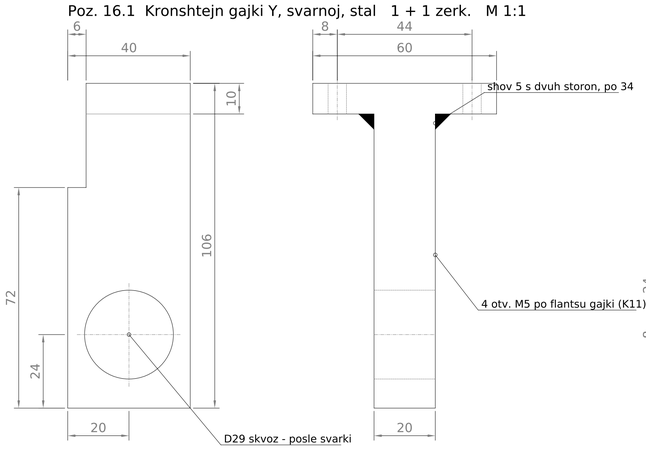
list 10; posle svarki
frezerovat niz, t >= 9,5

Poz. 15 Scheka, stal 6 4 sht. M 1:1



niz - k plite 8: razdelka snaruzhi 30x4, shov katet 6;
pered - sleva; t = 6

Poz. 16.1 Kronshtejn gajki Y, svarnoj, stal 1 + 1 zerk. M 1:1

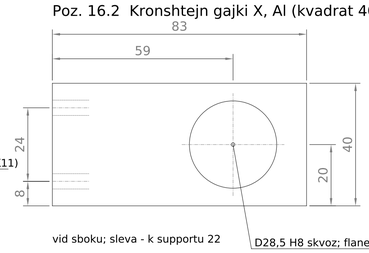


vid speredi

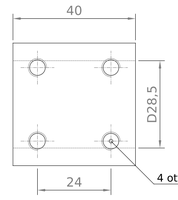
vid sboku (pered - sleva)

4 otv. M6 skvoz v polke (K12): X 16; 34, Y 8; 52.
Zagotovki St3: stojka - polosa 40x20, L=96 (stupenka 6x24 vypilit do svarki); polka - polosa 60x12, L=34.
Posle svarki frezerovat: verh polki v t=10, ploskostnost 0,05; zadnyuyu gran (pod flanets) - perpendikulyarno verhu 0,05/40, t >= 18; D29 - os perpendikulyarna zadnej grani; zatem rezby. Grunt/kraska, krome privalochnyh poverhnostej.

Poz. 16.2 Kronshtejn gajki X, Al (kvadrat 40x40) 1 sht. M 1:1



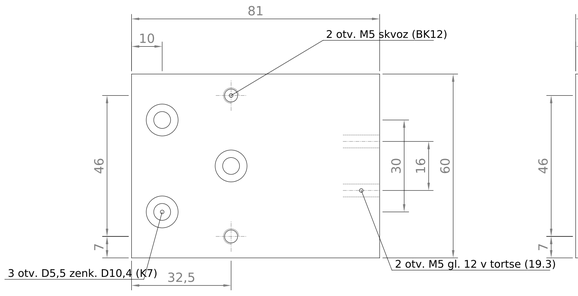
vid sboku; sleva - k supportu 22



vid na gran k supportu

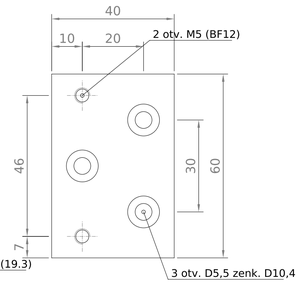
Zagotovka: kvadrat Al 40x40 (AD31T1 / D16T), L 86;
dopuskaetsya paket 4 plit Al 10 (83x40), styanut vintami flantsa M5 naskvoz, D28,5 - v sbore.
Tortsy frezerovat v razmer 83; gran k supportu i gran pod flanets - perpendikulyarno osi D28,5 <= 0,05.
Flanets gajki - lyskami vverh/vniz; M5 - po flantsu.

Poz. 19.1 Plita BK12 X, Al 10 1 sht. M 1:1



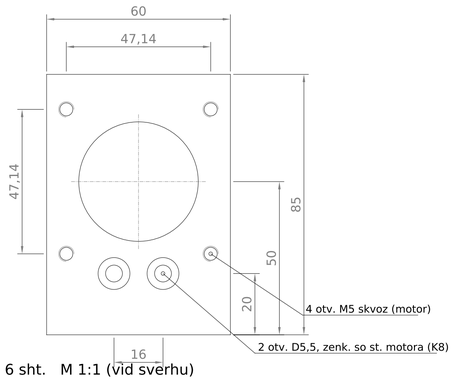
BK12 os - po u = 32,5; levyj kraj - k tsentru

Poz. 19.2 Plita BF12 X, Al 10 1 sht.

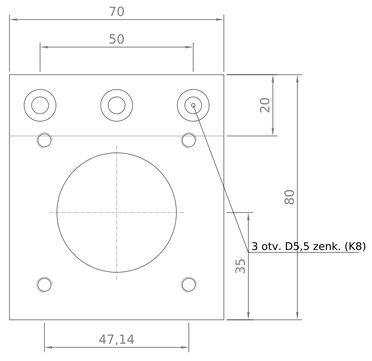


levyj kraj - torets balki

Poz. 19.3 Plita motora X, Al 10 1 sht. (D39 H11 - poyasok motora D38,1)

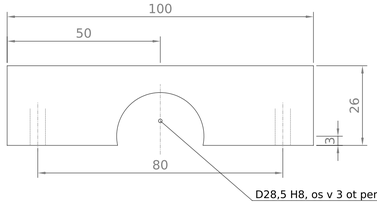


Poz. 19.4 Plita motora Z, stal 6 1 sht.

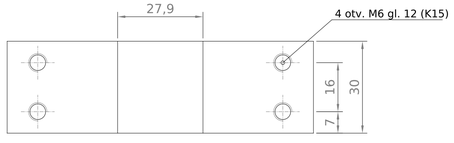


D39 H11; 4 otv. M5 (motor)
tonkaya - gran supporta 22

Poz. 16.3 Kronshtejn gajki Z, Al 1 sht. M 1:1

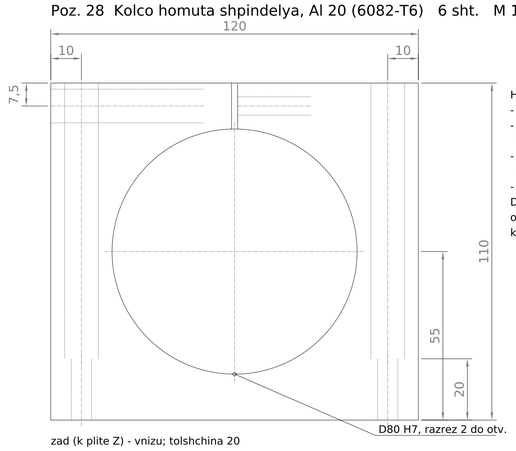


vid sverhu; pered (k plite Z) - vnizu



vid speredi; 3 otv. M5 sverhu - po flantsu (K11)

Poz. 28 Kolco homuta shpindelya, Al 20 (6082-T6) 6 sht. M 1:1 (vid sverhu)



zad (k plite Z) - vnizu; tolshchina 20

Homut = 3 kolca. V kazhdom kolce:
- razrez 2 do otv. styazhki;
- 2 otv. D6,6, tsek. D11 do 20 ot zadnej grani (K13);
- styazhka: D6,6, tsek. D11 sleva / M6 sprava (K14);
- vse otverstiya - po seredine (Z 10).

D80 H7 rastachivat v pakete 3 kolec odnogo homuta (styanut strubcinoj), kolca zamarkirovat.

Portal. Detali		
Spec. v2.2. Sborka - list 1	Masshtab 1:1	List 3 / 3